Prediction of Pump As Turbine (PAT) Performance

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1. Purpose

This document defines the procedure for preliminary selection of centrifugal pumps to be operated as turbine (PAT) or hydraulic power recovery turbine (HPRT).

2. Scope

The selection of a pump that will work as a turbine, can be based on a lot of different theoretical or experimental data based methods (Brkich, Javia, Ikeguchi, House, Schiavello, Stepanoff, Sharma, Gulich, Grover, Hergt, Alatorre, Barbarelli, Ventrone, Chapallaz).

The internally developed procedure follows the Javia's one [1] based on experimental coefficients. This method is also the most immediate and simple, with good matching between theoretical and tested curve (see technical paper "*Test of a Multistage Hydraulic Power Recovery Turbine*" – G. Agrati – Pompe Gabbioneta – Internal Proceedings '98 and master thesis "*Pumps as turbines: a study of the performance prediction methods*" – B. Rota).

This procedure is applicable to all pumps.

3. Acronyms

PAT: Pump As Turbine

HPRT: Hydraulic Power Recovery Turbine

BEP: Best Efficiency Point

CFD: Computational Fluid Dynamics

4. General Procedure – Description

The method can be described as follows:

- Evaluate turbine specific speed:

$$N_{St} = \frac{RPM_t \sqrt{Q_t}}{H_t^{0.75}}$$

Note: Q_t flow to be indicated per impeller entry; H_t head per stage
Multiply by 51.64 to obtain units of measure adopted in the graphs (conversion from SI to Imperial Units)

- From the graph of Figure 1 based on experimental data, evaluate flow coefficient C_q corresponding to N_{St} and maximum efficiency
- From another experimental curve evaluate head coefficient C_h corresponding to N_{St} and previous C_q value (see Figure 2)
- Evaluate service condition of the pump with the following expressions:

$$Q_p = \frac{Q_t}{C_q} \qquad H_p = \frac{H_t}{C_h}$$

where:

Q_p = pump capacity [m^3/h]

Q_t = turbine capacity [m^3/h]

C_q = flow coefficient []

H_p = pump head [m]

H_t = turbine head [m]

C_h = head coefficient []

- Select a pump having these values at best efficiency point of the pump (or closest possible to BEP and better if rated diameter is as near as possible to the maximum diameter)
- Finally, using the same experimental curves, draw the performance curves of the turbine on the basis of best efficiency point characteristic of the pump (capacity, head and efficiency).

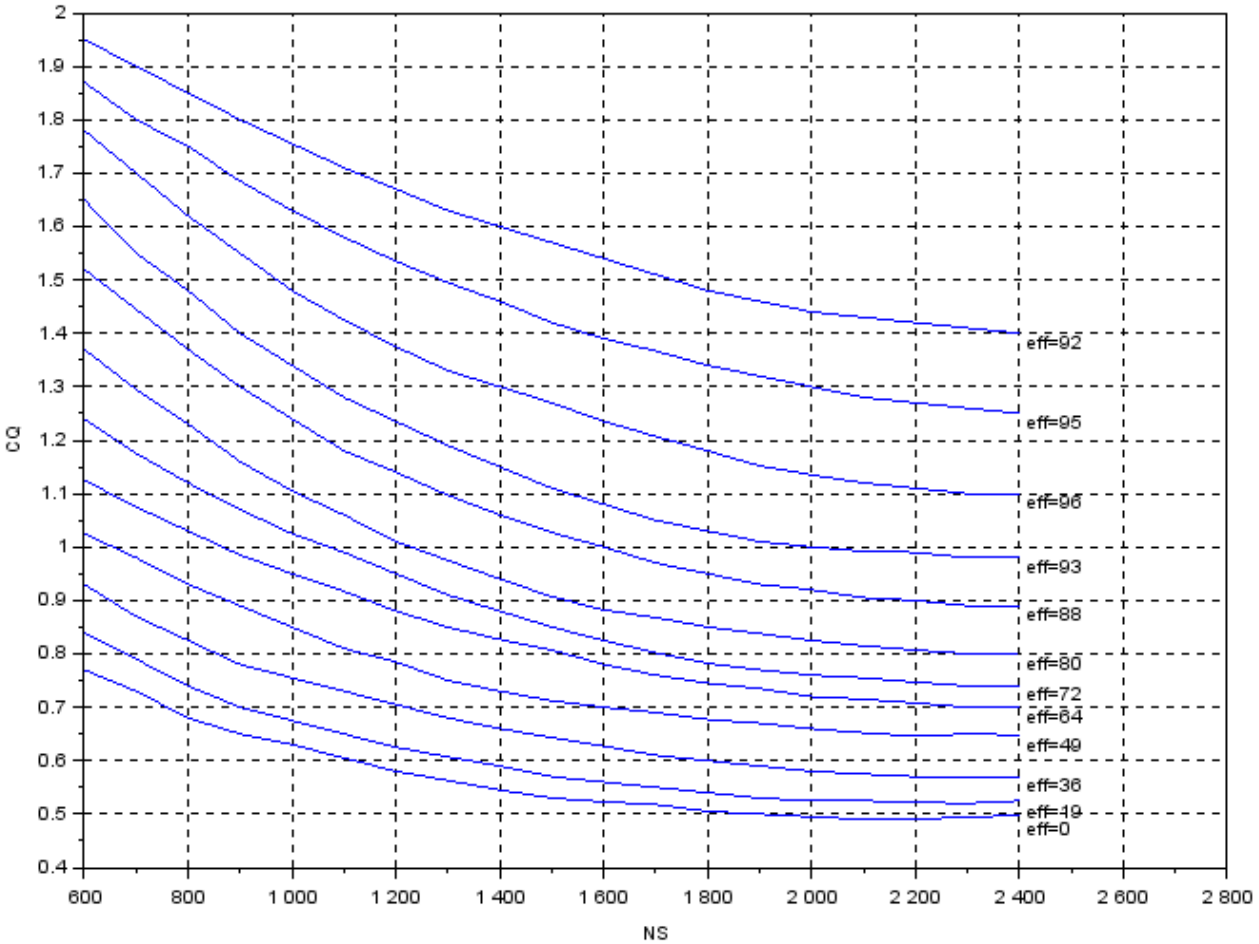

Figure 1. C_q Coefficients

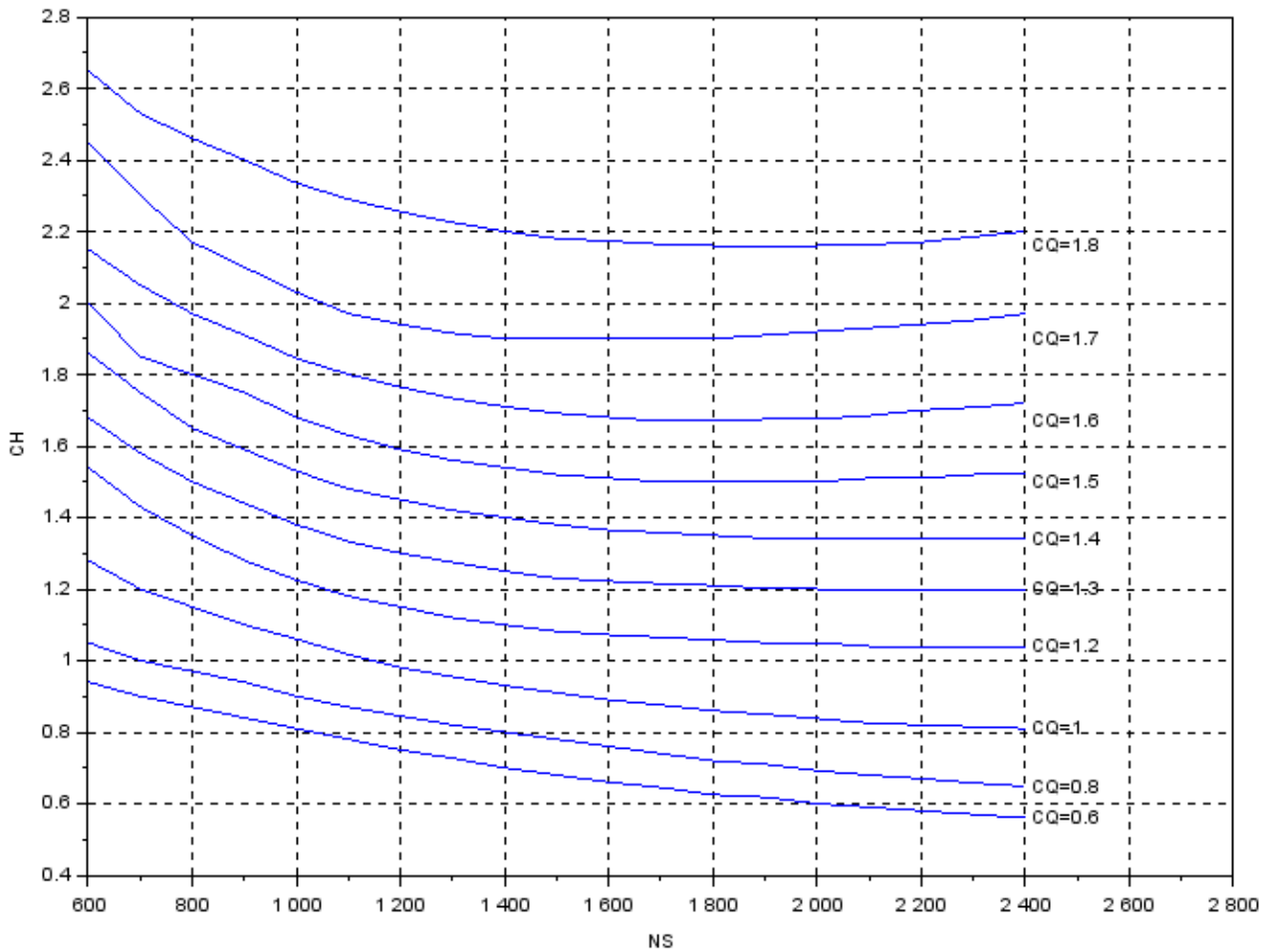

Figure 2. C_h Coefficients

Note:

The above graphs are directly obtained from the before mentioned bibliography.

5. Methodology, analysis and interpretation of results

Using Javia's method as described in the above section to select a PAT, it is important to pick up a pump with rated diameter the nearest possible to the maximum one. In fact, there shall not be too much space between impeller exit diameter (entry diameter under turbine mode) and cutwater so that the fluid is well guided from the volute towards impeller blades.

Moreover, rated point as pump shall be as close as possible to pump BEP otherwise correlation has more uncertainty and efficiency is not maximized.

Finally, it is important to evaluate the results with comparison to Gülich's method based on statistical correlation and described in Chapter 12 of *Centrifugal Pumps* book (see *Table 12.1 Turbine characteristics*).

6. Acceptance Criteria

This procedure allows to select a pump as turbine and issue a performance curve for proposal. During detailed engineering stage it will be necessary to carry out a CFD analysis in order to confirm the pump selection and optimize the hydraulics, if necessary.

7. Conclusions

Javia's method can be applied for selecting preliminary pump and issuing a proposal curve. Other selection and predictive methods such as Güilich, Sharma, Stephanov, etc. can be applied for comparison, but then there is necessity to assess pump performance and decide for possible hydraulic optimization through a CFD analysis.

8. References

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9. APPENDIX

9.1. Scilab Script

To obtain the preliminary performance curve of the pump as turbine, a tool developed internally within *Scilab* environment is available.

The software has a GUI so after launching the script called “HPRT_selector_rev3_GUI_update.sce” attached to this document (it is necessary to have *Scilab* installed on local PC and it is downloadable from https://www.scilab.org/download/scilab-2024.0.0), the user will be requested to indicate the performance as turbine (see Figure 3).

Then clicking the button “calculate” the tool gives back on another window the performance of the preliminary corresponding pump to be selected (Figure 4).

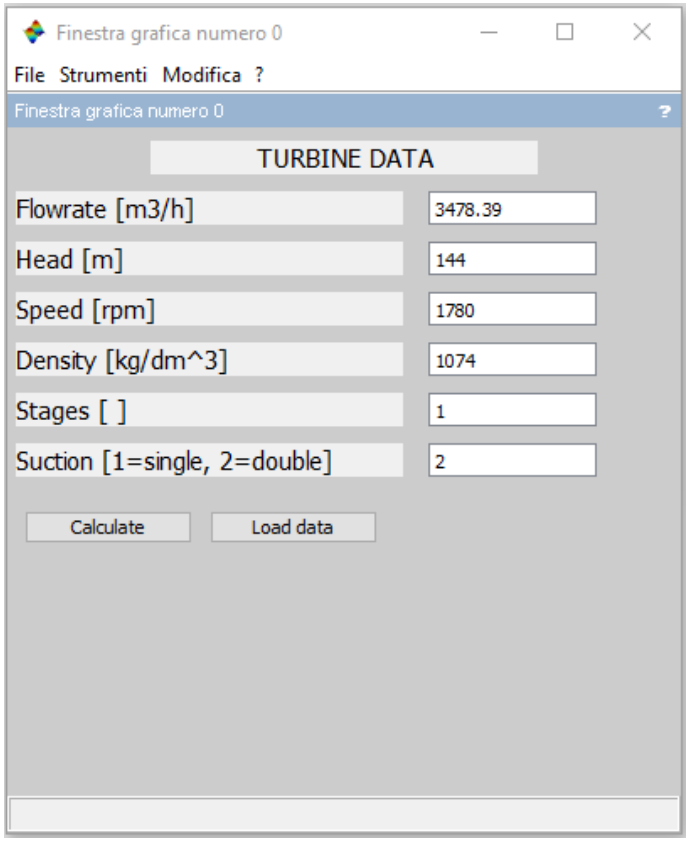

Figure 3. Input Mask for Turbine Data

Once a suitable pump is selected (rated flow very near to the BEP and rated diameter almost at maximum diameter), it is necessary to insert its efficiency and 100 for the “Actual % of pump bep” field. This functionality to take into account for selection of pump rated flow not at BEP (or very near to it) is still under development.

It is now possible to click the “continue” button and generate the pump performance curve as turbine. Some fields of the output format for the curve can be edited indicating the other project data (see Figure 5).

The selection process will probably be iterative because it is possible to regulate the speed, to change the number of stage and type of impeller (single or double suction) in order to find a suitable pump to be operated as turbine, of course considering also project constrains. For

example, if the turbine will drive a main pump, the speed may be fixed by the driven equipment in order to avoid the use of a gearbox.

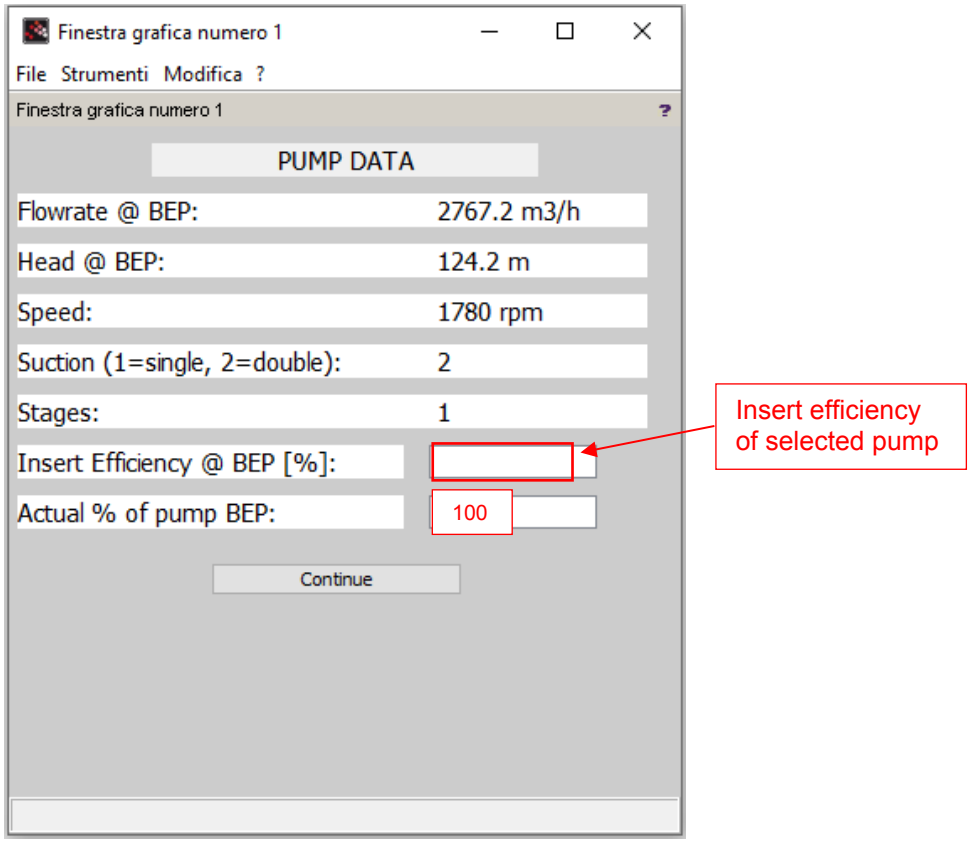

Figure 4. Output Data for Functioning as Pump

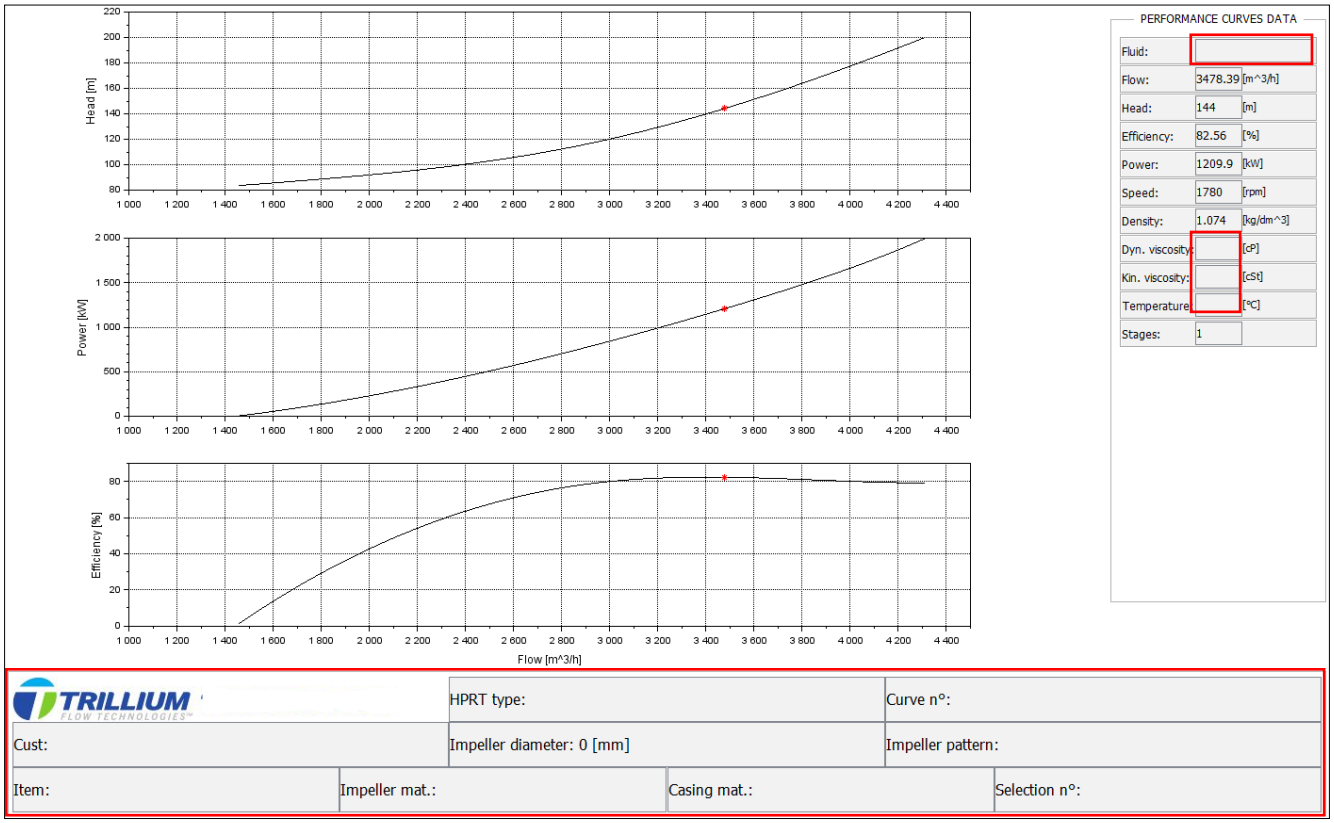

Figure 5. Example of Pump Performance Curve Output for Operation as Hydraulic Power Recovery Turbine